IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-013

1. Topic (12 words max)

A Centralized System for Real-Time Farm Animal Disease Reporting

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	R26-SE-013
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/gravity of the Problem

The livestock industry is a cornerstone of Sri Lanka's economy and national food security. Currently, the Department of Animal Production and Health relies on a manual, fragmented, and paper-based reporting pipeline where data is aggregated on a weekly or monthly basis. This "data lag" leaves Farmers vulnerable to sudden asset loss, Field Veterinary Officers without real-time diagnostic support, and National Directors unable to make rapid, informed decisions during fast-moving outbreaks like Foot and Mouth Disease.

Impact of the Problem

The impact is primarily Economic, as delayed responses lead to production losses, mass culling, and trade restrictions that devastate rural incomes. Socially, it threatens national food security by disrupting the supply of essential proteins like milk and eggs. Ethically and environmentally, the inability to track zoonotic diseases poses a severe public health risk, as unchecked animal pathogens can cross over to humans.

- **Economic**: Outbreaks cause devastating production losses leading to severe impacts on rural livelihoods and national trade.
- **Social & Public Health:** Delays in detection threaten the national protein supply (milk, meat, eggs) and increase the risk of zoonotic diseases jumping to humans.
- **Technical:** Existing surveillance lacks integrated, real-time "War Room" visualization and intelligent triage layers, creating a "skill gap" that delays official response.
- **Ethical & Environmental:** Inefficient monitoring often leads to late-stage reactive treatments, increasing the risk of antimicrobial resistance.

FoC Research Cluster and Industry Vertical/s

This research primarily focuses on Software Engineering (SE) to ensure the development of a robust, scalable, and integrated digital ecosystem. While the system leans into the Centre of Excellence for Artificial Intelligence (CEAI) cluster for its core diagnostic and predictive intelligence, the overarching goal is the engineering of a reliable national reporting platform.

Attainment of Sustainable Development Goals (SDGs)

The project directly addresses SDG 2 (Zero Hunger) by safeguarding national food production, SDG 3 (Good Health and Well-being) by controlling zoonotic disease vectors, and SDG 8 (Decent Work and Economic Growth) by protecting the economic stability of the farming community.

References (Supporting the Research Problem)

1. **Department of Animal Production and Health (DAPH). (2023).** *Annual Performance Report.* (Addresses: Manual reporting gaps and current livestock sector status in Sri Lanka).

   Link : https://daph.gov.lk/downloads/daph-publications

2. **Abeynayake, P., & Stephen, C. (2012).** *"Gap Assessment of Animal Health Legislation in Sri Lanka for Emerging Infectious Disease Preparedness."* Outlook on Agriculture. (Addresses: Legal and institutional surveillance deficiencies).

   Link : https://www.researchgate.net/publication/264389517_Gap_Assessment_of_Animal_Health_Legislation_in_Sri_Lanka_for_Emerging_Infectious_Disease_Preparedness

3. **Frontiers in Sustainable Food Systems. (2025).** *"Livestock and poultry production in Sri Lanka: Industry resilience and disease management."* (Addresses: Economic impact and productivity losses).

   Link : Frontiers | Livestock and poultry production in Sri Lanka: challenges and strategies for climate-resilient food security

4. **World Organisation for Animal Health (WOAH). (2025).** *State of the World's Animal Health.* (Addresses: Zoonotic risks and the role of diagnostic tools in trade transparency).

   Link : Statement on recent lumpy skin disease outbreaks in Europe - WOAH - World Organisation for Animal Health

5. **International Livestock Research Institute (ILRI). (2016).** *Livestock and the Sustainable Development Goals.* (Addresses: SDG 2, 3, and 8 alignment).

   Link: https://www.ilri.org/

6. **Ocholla, P., et al. (2024).** *"Public Computer Vision Datasets for Precision Livestock Farming."* arXiv:2406.10628. (Addresses: Technical gaps in field-based data collection).

   Link : https://arxiv.org/abs/2406.10628

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products/Systems

Globally, animal disease surveillance is dominated by high-level administrative platforms such as the World Animal Health Information System (WAHIS), which focus on macro-level, retrospective data reporting for international policy and trade. In the commercial sector, Precision Livestock Farming (PLF) systems utilize IoT sensors for individual animal tracking; however, these are typically expensive, closed-source, and prioritize production metrics (e.g., milk yield) over real-time clinical diagnostic surveillance in rural, low-resource settings.

Summary of Existing Research

Current academic research extensively explores Convolutional Neural Networks (CNNs) for detecting specific livestock pathologies, such as Lumpy Skin Disease, often achieving high accuracy in controlled laboratory settings. Additionally, time-series models like LSTM are increasingly applied to epidemiological forecasting to predict future outbreaks. However, these studies often operate in "silos," focusing on single-disease detection or theoretical prediction without integrating multi-modal inputs into a unified, field-deployable ecosystem.

Gaps/Limitations Identified

AI-Powered Smart Diagnosis system:

- **Domain Shift:** Models trained on "clean" lab datasets often fail in rural field conditions due to environmental noise, such as mud and varying lighting.
- **Uncertainty Estimation Gap:** Existing "black-box" models provide confident diagnoses even for unknown pathologies, lacking the ability to flag ambiguity for professional veterinary review.
- **Rare Disease Data Scarcity:** A significant imbalance exists in annotated datasets, where rare but high-risk tropical diseases are underrepresented, leading to biased models.
- **Interspecies Morphological Variance:** Most diagnostic research fails to account for skin/hide differences across species (e.g., cattle vs. buffalo), resulting in poor diagnostic generalization in multi-species environments.

AI-Driven Wellness & Management Standardization

- **Subjectivity in Physical Assessment:** Manual Body Condition Scoring (BCS) is now subjective, which results in the provision of inconsistent health records in various farms.
- **Reactive Monitoring Gap:** Current systems are currently geared towards the detection of disease once it has become symptomatic and not the sub-clinical signs such as a reduction in feed and water intake that are pre-illness events.
- **Lack of Tropical Health Benchmarks:** The majority of wellness data were done with Western breeds and thus the benchmarking is not accurate on Sri Lankan cattle and buffalo.
- **Image Processing Robustness Gap:** Current automated scoring systems are typically not applicable to rural setting because of image noise (mud, background clutter) and changing lighting, which hides the skeletal features (ribs and spine) that must be accurately identified to assess the patient's health.

Seasonal Disease Forecasting System:

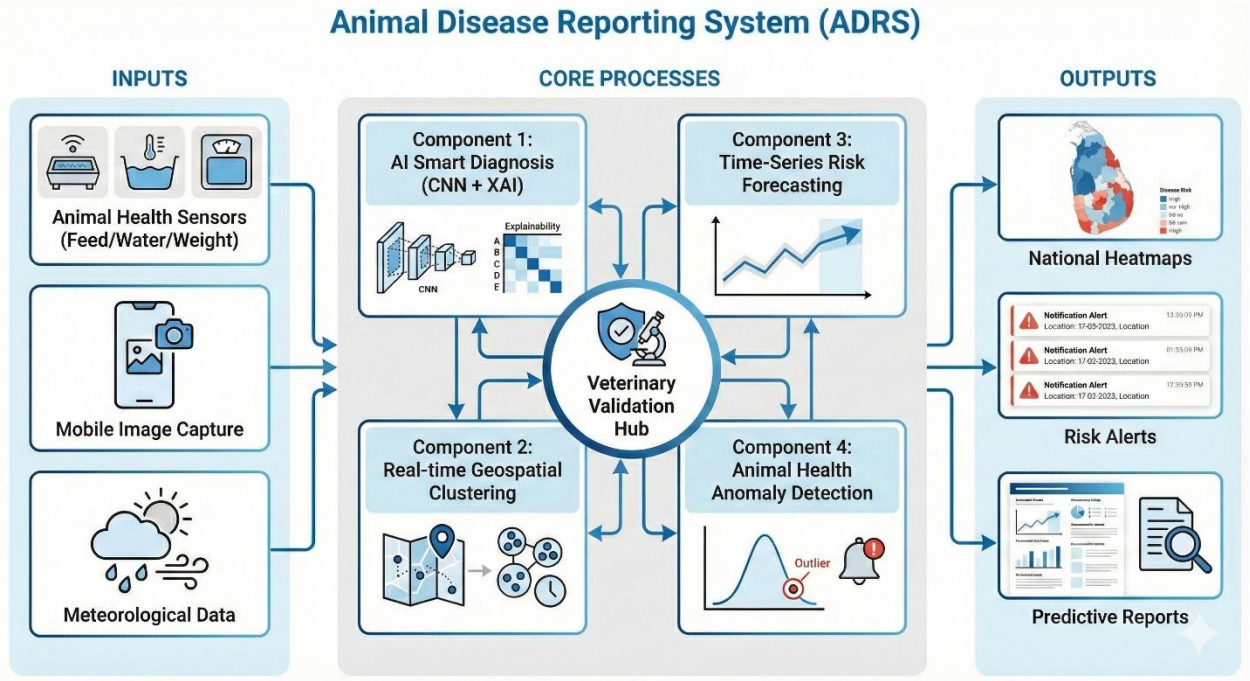
- **Linearity-Complexity Mismatch:** Conventional statistical forecasting models assume of linear and constant seasonal behavior, which are not effective in suddenly, non-linear outbreak spikes that occur frequently with livestock diseases like Black Quarter.
- **Sparsity in historical Data:** Data on livestock diseases in Sri Lanka has a limited duration and frequency, and many diseases have longer zero-reporting periods, which causes the imbalance in data and unreliable training of deep learning models.
- **Omission of Environmental Variables:** Current methods of forecasting are mostly based on the number of cases in the past and do not include the important environmental factors like soil moisture and the intensity of rainfall, which are the biological causes of disease outbreak.

- **Model Explainability Deficit:** The majority of advanced forecasting models are treated as black boxes, which are models making risk predictions without explanations, thus restricting trust and viable application by veterinary professionals and decision-makers.

Real-Time Geospatial Visualization:

- **Temporal Data Lag (Retrospective Reporting**): The existing veterinary reporting workflow is based on manual, paper-based aggregation, which means that maps show past outbreaks but not the current situation, and thus response cannot be taken immediately.
- **Limitations of Static Visualization:** The current systems make use of simple statical point markers to display the location of cases, and they cannot display the important dynamic property of disease density, speed of spreading, or directionality.
- **Informational Noise:** The existing systems do not have the ability to differentiate between unconfirmed rumors and those that are confirmed by trained professionals, creating the so-called false alarms and wasting scarce government resources.
- **Loss of Goods to Environmental Triggers:** The traditional disease mapping fails to incorporate the environmental drivers of wind direction, rainfall or water flow and therefore it cannot be possible to comprehend the how and why of an exact spread path.
- **Absence of Actionable Decision Support:** Decision-makers do not have an interface to transform complex spatial data into explicit mathematical suggestions concerning containment zones or vaccination priorities, so manual approximation is used.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



How the System Addresses Identified Gaps

The Animal Disease Reporting System (ADRS) is designed to transition Sri Lanka's veterinary surveillance from a manual, reactive process to a proactive, "Human-in-the-Loop" (HITL) digital ecosystem. It specifically addresses the following gaps:

AI-Powered Smart Diagnosis system:

- **Environmental Robustness & Domain Shift:** The system incorporates a robust pre-processing layer for mobile image capture to standardize noise (e.g., lighting, mud) before diagnostic analysis.

- **Trust & Uncertainty Estimation:** By utilizing Explainable AI (XAI) and Grad-CAM visualizations, the system highlights precisely which pathological features are being analyzed. This allows the Veterinary Validation Hub to serve as a critical guardrail, preventing overconfident AI "silent failures" from corrupting national data.

- **Data Scarcity for Rare Diseases:** The AI models utilize advanced data augmentation to improve detection sensitivity for high-risk, low-incidence diseases (e.g., HPAI, Brucellosis).

- **Interspecies Variance:** The diagnostic component is species-aware, adapting its feature extraction to account for the morphological hide differences between cattle, buffalo, and goats.

AI-Driven Wellness & Management Standardization

- **Objective Wellness Standardization:** The system is a Computer Vision initiative that allows automation of BCS and removes human error with the goal of standardizing a digital health indicator.
- **Proactive Management Triage:** The system makes it possible to intervene in advance of clinical manifestations (i.e. water/feed consumption anomalies) by analyzing them and addressing them before they manifest themselves in the patient.
- **Localized Benchmarking:** The AI models are adapted to the local environment where Sri Lankan tropical breeds are located so that management instructions are relevant to local conditions.
- **Adaptive Image Processing Pipeline:** The system operates based on a multi-step preprocessing layer that incorporates **Contrast-Limited Adaptive Histogram Equalization (CLAHE)** to amplify the presence of bone-shadow on the photos and Gaussian noise suppressions to maintain the presence of clear features on the extracted image products under normal field conditions regarding mobile photos.

Seasonal Disease Forecasting System:

- **Comparative and Hybrid Forecasting Framework**: The system evaluates multiple forecasting approaches, including statistical time-series models and machine learning-based models, to identify the most suitable method for capturing both regular seasonal disease patterns and irregular non-linear outbreak behavior. Hybrid modeling strategies (ARIMA-LSTM architecture) are explored where appropriate, based on empirical performance rather than prior assumptions.
- **Mitigation to Data Scarcity:** The system deals with insufficient data in the past using synthetic **data augmentation** and **transfer learning** of similar tropical areas, and weekly data aggregation enhances the quantity of effective training samples.
- **Integration of Environmental Data:** The forecasting model utilizes major aspects of the environment including satellite-derived soil moisture, rainfall and temperature which then facilitate prediction based on the existing climatic conditions which cause livestock disease outbreaks.
- **Explainable Prediction Mechanism:** Explainable AI (XAI) methods are combined to explain the results of every risk forecast in a manner that is easy to understand and

understand by human, so that
veterinary officers can learn, trust and come up with the predictions that are easy to understand and verify before taking any preventative measures.

Real-Time Geospatial Visualization:

- **Live tactical Intelligence:** With the help of WebSockets, the system is no longer what happens after the event, but rather it reflects the current situation, and the map is automatically updated as a case is confirmed.
- **High-Fidelity Data Standard:** The system is sensitive to the issue of informational noise, by making only the so-called Vet-Confirmed cases visible, such that the War Room dashboard can be a 100% trustworthy source of official policy decisions.
- **Spread Velocity and directional mapping:** The system contrasts with fixed tracker operations; unlike the former, it implements the Spatio-Temporal Analysis to determine the speed (km/day) and direction of the outbreak, which results in dynamic arrows of the vectors indicating the direction in which the disease will follow.
- **Integration of Environmental Triggers:** The system integrates the clusters of outbreaks with live environmental data (e.g., wind speed, soil moisture) to be able to visualize the actual triggers that promote the spread so that scientifically justified interventions can be made.
- **Automated Actionable Recommendations:** The system transforms raw spatial data into strategic action through automatic generation of Quarantine Boundaries and Vaccination Priority Areas, which is a calculated risk level with a cluster density.

System Perspective (Input, Process, Output)

Input:

- **Animal Health Metrics:** Real-time data from farm-level sensors monitoring feed intake, water consumption, and weight.
- **Visual Data:** High-resolution mobile images of visible symptoms (e.g., skin lesions, mucosal abnormalities) captured by field officers.
- **Environmental Context:** Meteorological data including monsoon patterns and temperature changes to inform epidemiological risk.
- **Animal Profile Metadata:** The non-dynamic data such as breed type, actual age, sex and current production phase (e.g., lactation day or growth cycle) is used to build the appropriate baseline of what is referred to as optimum.
- **Past Wellness Scores:** Past data of the individual animal to enable the AI to trace long-term health trends as opposed to individual snapshots of the animal.
- **Disease Data:** Past Table of historic outbreaks by year, place, disease and number of cases.

- **Weather Wizardry:** Precipitation, temperature, moisture, and monsoons.
- **Satellite Data:** Vegetation indicators and soil moisture.
- **Supporting Data:** Density and vaccination coverage of livestock population.
- **High-Fidelity Outbreak Metadata:** Latitude and longitude location of the Vet-Confirmed cases by the type of disease and severity.
- **Temporal Synchronization Data:** Accurate timestamps of case validation to be used in determining the real-time movement and acceleration of outbreaks.
- **Geographic Geospatial Boundaries:** GeoJSON /Shapefile data of Sri Lankan Provinces, Districts, and Divisional Secretariat (DS) boundaries to be able to map the regions.
- **Predictive Risk Vectors:** Future infection potentials (through the use of the Spread Prediction model) in the form of geocoded probability scores.

Process:

- **Anomaly Detection:** Analyzing biometric data to identify sub-clinical health drops before visible symptoms appear.
- **Intelligent Diagnosis (CNN + XAI):** A Convolutional Neural Network analyzes images for differential diagnosis, providing visual heatmaps for verification.
- **Human-in-the-Loop Validation:** All AI-flagged "High Risk" cases must be professionally reviewed and confirmed by a veterinarian.
- **Geospatial & Predictive Analytics:** Only *confirmed* cases are fed into geospatial clustering algorithms and Time-Series (ARIMA/LSTM) models.
- **BCS Feature Extraction:** The CNN model finds and quantifies fat cover on the ribs, spine and tail-head over anatomical features of the photos such as the ribs.
- **Normalization of data:** When raw feed and water intake are transformed into relative intake percentage depending on the present body weight of the animal and the temperature of the environment.
- **Pre-processing and Noise Reduction:** Farms Standard mobile photo images have to contend with an environmental noise (mud on the animal, low sun, or a busy background). A Gaussian Blurring method is used on the system to eliminate high-frequency noise, as well as the Contrast -Limited Adaptive Histogram Equalization (CLAHE). The importance of this particular technique is that the shadows of the structure of the bones of animals (ribs and spine) are higher so that even in the bad light one can see the landmarks.
- **Multi-Modal Integration:** The final image-derived score is not used alone. It is "fused" with numerical data (water/feed intake) through a **Feature Fusion Layer**. This ensures that if an animal *looks* healthy but is *drinking* 50% less than usual, the system can flag a "Sub-clinical Wellness Warning.

- **Anatomical Landmark Localization:**
  This is the core "Computer Vision" task. The system identifies specific **Keypoints** (Hooks, Pins, and Tail-head). By using a **Regression-based CNN**, the system measures the "pixel-distance" and "curvature angle" between these points. A sharp angle indicates a thin animal (low fat), while a rounded curve indicates a healthy or overweight animal.
- **Cohort Peer Analysis:** Raw figures of a single animal are compared to the average of the so-called peer group (of the same breed/age) in that farm to determine outliers.
- **Prescriptive Mapping Logic**: A decision-tree process that contains the identified health gaps (e.g., "Low BCS"), and identifies the managed protocols (e.g., "Add protein concentrate by 0.5kg") that are already pre-vetted.
- **Data Engineering** - Gets data out of PDF reports, removes missing values and eliminates anomalies, Matches disease and climate data against time and place.
- **Feature Engineering** - Produces lag characteristics (rainfall and cases of the past weeks), Calculates rolling averages and climate anomalies, Identifies severe weather conditions.
- **Modeling** - The long-term seasonal trends of ARIMA models, LSTM acquires nonlinear and complicated outbreak behavior, the combination of both models is in final prediction.
- **Evaluation & Explainability** - Measures accuracy based on conventional measures, Produces descriptions about every prediction.
- **Unsupervised Geospatial Clustering (DBSCAN**): This is a machine learning algorithm that automatically clusters individual confirmed cases in spatial clusters to determine the emerging Hotspots in terms of proximity and density.
- **Kernel Density Estimation (KDE):** This is a process that involves the calculation of smooth intensity gradients on a geographic area to generate heatmaps, which represent the visualization of the weight of the disease burden.
- **Spatio- Temporal Spread Analysis:** A method that examines the change in the centroid of the clusters with time to have a mathematical calculation of the Spread Velocity and directional movement of an outbreak.
- **Automated Containment Logic:** This is a geometric processing layer (calculates high-risk perimeters around clusters) to propose Quarantine Boundaries and Surveillance Buffer Zones automatically.
- **Real-Time Rendering Pipeline:** Live situational awareness is maintained by means of using WebSockets and special APIs to push validated data updates to the map interface, rendering the map without having to refresh the page manually.

Output:

- **National Heatmaps:** Dynamic visualizations of disease density and spread velocity for district directors.
- **Risk Alerts:** Automated, high-priority notifications for emerging "hotspots" to enable rapid resource deployment.
- **Predictive Reports:** Seasonal forecasts identifying high-risk regions weeks in advance for pre-emptive vaccination.
- **Individual Wellness Score Card:** This is a numerical index (0-100) that provides the farmer with an immediate form of a Health Grade of animal.
- **Visualizations:** Wellness Trends: Interactive charts displaying the change in Body Condition Score (BCS) and weight gain during weeks or months.
- **Pre- Symptomatic Management Alerts:** High-priority messages on the Sub-Clinical issues like a slight decrease in water consumption, but which has not yet been noticed by a vet but requires a management amendment.
- **Optimal Production Targets:** This is the projection of the possible milk or weight gain the animal may achieve provided the recommended instructions on management are adhered to.
- **Short-Term Seasonal Disease Forecast**. **(1–4 weeks):**
    - Confidence range of probability of disease.
    - Timely notification of vaccination plans.
- **Medium-Term Seasonal Disease Forecasts (1–3 months):**
    - Seasonal risk calendars and regional heatmaps.
- **Long-Term Seasonal Disease Insights (6–12 months):**
    - Trends in comparative risk over the years.
    - Projections of disease burden due to climate changes.
- **Interactive "War Room" Dashboard:** This is a centralized, live command interface to the District and Provincial Directors, giving them a 360-degree situational view of the national animal health situation.
- **Dynamic National Heatmaps:** Interactive visual layers that show the intensity of the disease, with color scales (Red-to-Green) of high-density infection areas and safe areas.
- **Automated Hotspot Alerts:** A hotspot is an instant, location based notification of a new high-density cluster (hotspot), mathematically determined within an administrative boundary.
- **Strategic Resource Deployment Overlays**: Visual indicators of high-sensitive areas where vaccination teams, medical supplies, and restrictions on movement should be immediately implemented.

- **Zonal Containment Visuals:**
  Perimeters digitally illustrated on the map to designate certain areas where one-time quarantine and movement control of animals is to be enforced.

Gaining Essential Specialized Domain Expertise

The project team will acquire specialized domain expertise in veterinary pathology and livestock epidemiology through a multi-faceted approach. First, we will conduct an intensive review of peer-reviewed research papers and veterinary clinical guidelines to understand the morphological features of tropical livestock diseases. To bridge the gap between computer science and veterinary medicine, we will utilize AI-driven research tools to demystify complex medical terminology and pathology reports. Furthermore, team members will enroll in a final-year elective, such as " Machine Learning" or " Image Understanding and Processing" which closely aligns with the technical demands of the Smart Diagnosis component. Most importantly, we have established a relationship with a professional client in the veterinary field who will provide expert guidance, clinical insights, and validation of the diagnostic criteria used in our models.

Gaining Technical Knowledge and Skills

Technical proficiency in building the ADRS will be developed through hands-on experience with industry-standard frameworks like TensorFlow for deep learning. We will leverage specialized technical resources, including whitepapers on "Creating Veterinary Pathology AI Models" and webinars from digital pathology platforms to understand best practices in image annotation and model validation. Practical skills in geospatial analysis and time-series forecasting will be honed through online technical workshops and by analyzing open-source codebases for similar epidemiological projects.

Data Collection Methodology

Data for this project will be sourced using a hybrid strategy to ensure a robust and balanced dataset:

- **Public Repositories:** We will utilize established image datasets from platforms like Kaggle and Mendeley, specifically targeting skin disease datasets for initial model training.
- **Client/Primary Data:** Our client will provide primary clinical data, including historical outbreak records and symptom logs, to ground our models in the Sri Lankan context.
- **Meteorological Data:** Seasonal and weather variables will be sourced from open government weather portals to provide the necessary inputs for the forecasting component.

- **Performance Benchmarking Data:** The Department of Animal Production and Health (DAPH) will supply us with the data on the Optimal Growth Curves and Standard Consumption Tables to establish the healthy standard of the AI.
- **BCS Image Training:** This will involve first training the model using the public repositories such as Mendeley or Kaggle on the body conditions of the animals and then using primary data of local farms to fine-tune.

Ethical Clearance

Yes, this project requires ethical clearance. Since the research involves the use of data related to diseased animals and potentially sensitive information from farm owners, we must adhere to the guiding principles of animal welfare and data privacy.

8. Objectives and Novelty

Main Objective To develop and implement a high-integrity, AI-driven diagnostic framework within a centralized animal disease reporting system that utilizes Computer Vision and Human-in-the-Loop validation to minimize data lag and ensure the accuracy of national-level disease surveillance in Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT21129544	*AI-Powered Smart Diagnosis (Computer Vision)*	*1. Develop a Multi-Class CNN model for the differential diagnosis of overlapping tropical livestock symptoms (e.g., LSD vs. Pseudo-LSD).* *2. Implement Explainable AI (XAI) using Grad-CAM to provide visual evidence for clinical verification.* *3. Build a "Firewall" Validation Workflow to prevent unconfirmed AI predictions from affecting national maps.*	*This system is designed to work on real-world farms by cleaning up messy or poor-quality photos (like those with mud or bad lighting) before analyzing them. It also understands the physical differences between cattle, buffalo, and goats, ensuring a more accurate diagnosis for each specific animal. To improve reliability, the AI recognizes when it is confused or lacks enough information and will flag those cases for a vet to check manually instead of making a guess. Additionally, it uses AI-created images to train for rare diseases, allowing the system to detect dangerous outbreaks even*

			when real-life photos are hard to find.
IT22221414	*Seasonal Disease Forecasting (Predictive Analytics)*	*1. Gather and process past outbreaks of livestock diseases in Sri Lanka.* *2. Combine seasonal and environmental factors like rain, temperature, and monsoon seasons and disease records.* *3. Compare time-series forecasting models (ARIMA and LSTM) to forecast disease outbreaks and detect them weeks before.* *4. Produce forecasts of the risk of the disease on a province-level to facilitate early intervention and vaccination strategies.* *5. Test the accuracy of the model with historical unseen data and use the performance to refine the predictions.*	*1.Introduces **proactive disease forecasting** rather than traditional reactive outbreak reporting in Sri Lankan veterinary surveillance.* *2. Combines **seasonal climate patterns with historical disease data**, which is rarely applied in local livestock health systems.* *3 Provides **early warning risk forecasts** (weeks ahead) to enable preventive vaccination and resource allocation.* *4.Tailored specifically for **tropical, monsoon-driven disease behavior**, unlike most existing models developed for temperate regions.*
IT22194558	*AI-Driven Wellness & Management Standardization*	*1. Develop a **Regression/ML model** to establish wellness benchmarks for Sri Lankan livestock.* *2 Develop an automated **Image Processing Pipeline** featuring*	*The primary innovation of this component lies in the development of a **Lightweight 2D Computer Vision Framework** specifically optimized for **Tropical Breed Morphology**. While*

		***Background Subtraction**, **Landmark Detection**, and a **CNN-Regression model** for objective Body Condition Scoring (BCS).* *3. Design a **Prescriptive Engine** to generate non-medical management instructions based on health anomalies.*	*existing research in Precision Livestock Farming (PLF) often relies on expensive 3D depth-sensing cameras or controlled laboratory lighting, this system introduces a robust **image-processing pipeline** capable of extracting high-fidelity health metrics from **unstructured, low-resolution mobile images**. By implementing **Anatomical Landmark Localization** tailored for the unique physical traits of Sri Lankan cattle (such as the thoracic hump of the Sahiwal breed), the system provides a standardized, objective **"Wellness Score"** that was previously only achievable through expert manual palpation. This democratizes high-end veterinary diagnostic tools for small-scale rural farmers, shifting the industry **away from mere reactive disease treatment and towards proactive, data-driven management standardization.***
IT22258908	*Real-Time Geospatial Visualization (Interactive Intelligence Map)*	***1.Geospatial Case Plotting:** Implement a high-resolution mapping interface to plot "Vet-Confirmed" outbreak cases using GPS coordinates, integrating both*	***1.Transition to Live Tactical Intelligence:** This component transforms national veterinary response from **Retrospective***

		historical and current data to identify long-term spatial patterns. ***2.Advanced Spatial Clustering & Risk Visualisation:*** *Develop and optimize* ***DBSCAN*** *and* ***Kernel Density Estimation (KDE)*** *algorithms to detect high-density hotspots and assign risk-based color-coding (Red/Orange/Yellow) based on cluster severity and outbreak density.* ***3.Environmental Trigger Correlation:*** *Integrate live data feeds from the Forecasting component (wind speed/direction, rainfall, water flow, and soil moisture) to analyze the "why and how" of disease movement through environmental correlation.* *4.****Spread Velocity & Directional Calculation:*** *Develop mathematical models to calculate the outbreak's* ***Spread Speed (km/day)*** *and trajectory, rendering dynamic directional vectors (arrows) on the map to visualize the predicted path of infection.*	***Reporting*** *(logging past events) to* ***Live Tactical Intelligence****, where only high-fidelity, "Vet-Confirmed" data is used to eliminate informational noise.* ***2.Environmental-Linked Spread Prediction:*** *A major innovation is the integration of* ***Environmental Triggers*** *(wind, water, soil) as active variables in spatial mapping. Unlike generic trackers, this system calculates the* ***Speed and Direction*** *of an outbreak based on environmental flow, allowing authorities to see where the disease is going before it arrives.* ***3.Automated Containment Precision:*** *The system introduces* ***Mathematical Actionable Recommendations****, where the map automatically generates precise quarantine coordinates and vaccination cordons. This allows non-technical officials to make high-*

		5.Actionable Recommendation Engine: *Create an automated overlay system that suggests preventive measures based on spatial analysis, including* ***Quarantine Boundaries*** *(High-Risk),* ***Vaccination Priority Areas*** *(Medium-High), and* ***Surveillance Zones*** *(Low-Medium).* *6.****Real-Time Data Rendering:*** *Utilize* ***WebSockets*** *to ensure the map, risk levels, and environmental overlays update instantaneously without manual refresh as soon as a case is confirmed.* ***7.System Validation & Scientific Justification:*** *Validate the accuracy of clustering and spread predictions by benchmarking against historical GPS outbreak datasets from the DAPH, ensuring research depth in risk-level modeling.*	*stakes containment decisions with scientific precision rather than manual estimation.*

9. **This part is to be filled by the Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor		Dilani	Lanugalage	
Co-Supervisor	Mr.	Eishan	Weerasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				